Integer Linear Programming

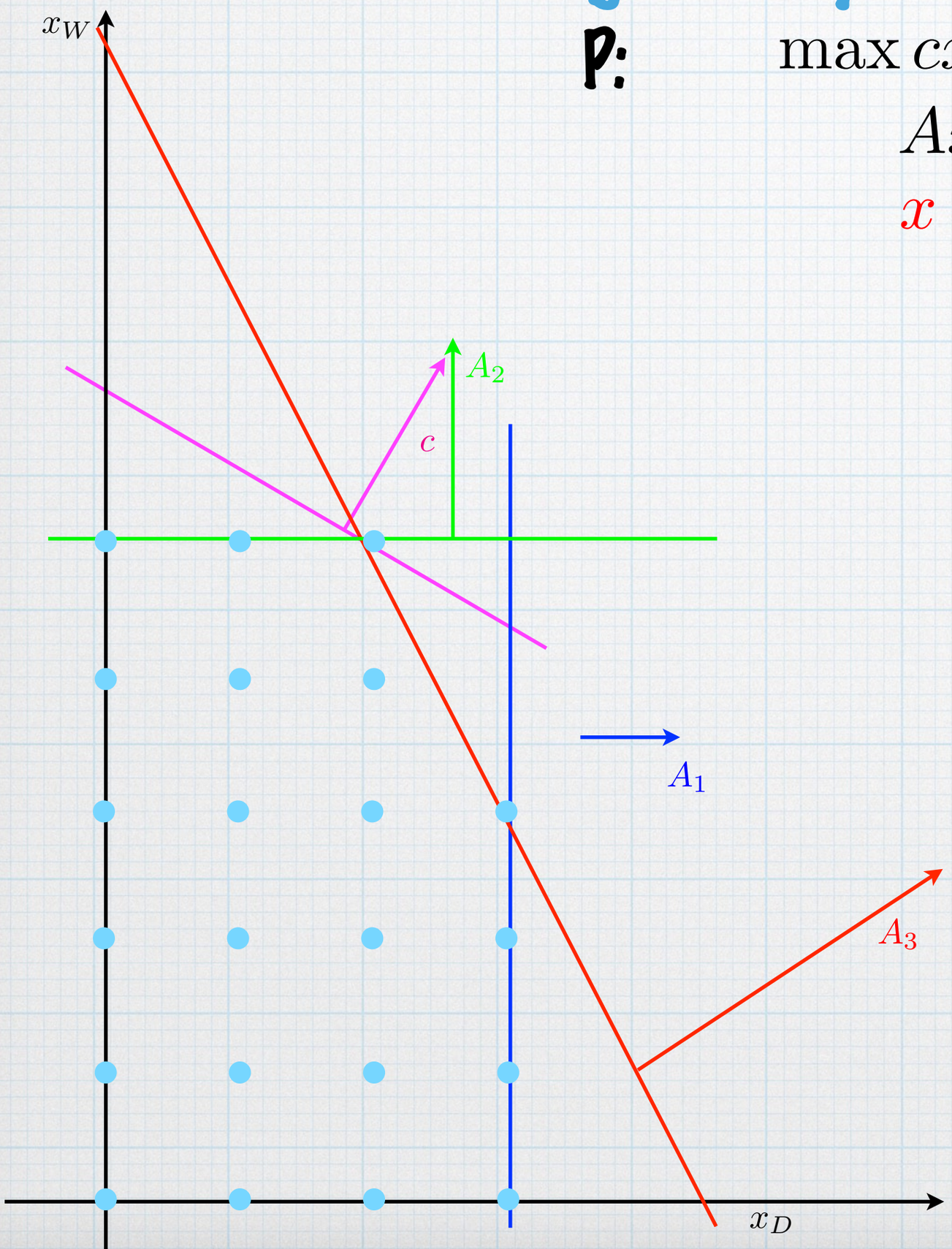
ILP challenges

* What happens when we add **integrality constraints** to a LP problem?
  * advantages
  * disadvantages
* When can we **ignore the integrality** constraints?
* Can we take advantage of the **LP techniques** to solve ILP problems?
* How can we speed up a **brute force enumeration**?

Link with your previous knowledge

* Remember the main properties of LP
  * optimum on a vertex
  * **basic solution optimality conditions**
* Brute force enumeration: how to enumerate all feasible solutions?
* Modeling techniques

Integrality constraints

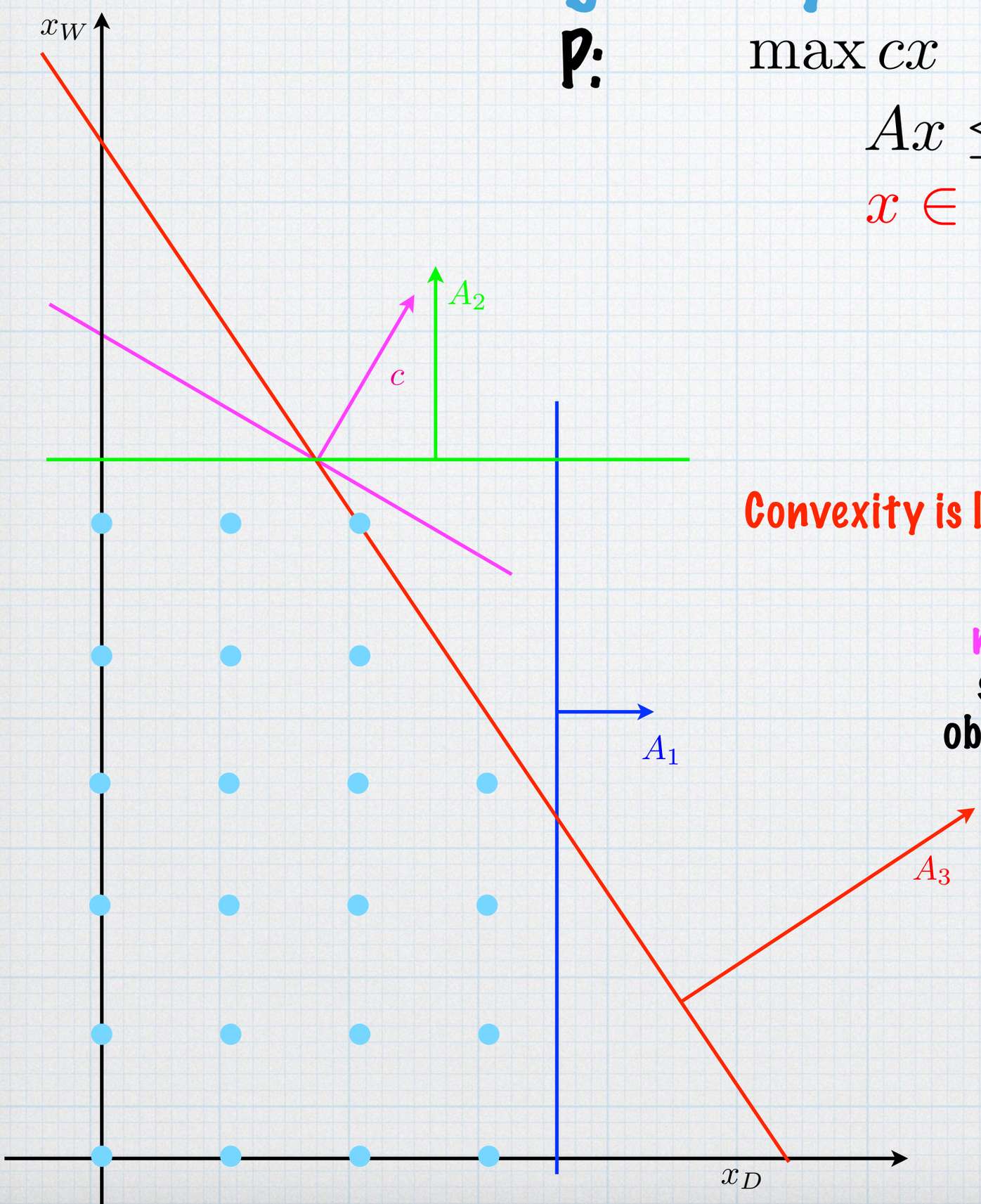


P:

$$\max cx$$
$$Ax \leq b$$
$$x \in \mathcal{Z}^n$$

$$\begin{aligned} \max 30x_D + 50x_W & \\ x_D & \leq 4 \\ 2x_W & \leq 12 \\ 3x_D + 2x_W & \leq 18 \\ -x_D & \leq 0 \\ -x_W & \leq 0 \\ x_D, x_W \in \mathcal{Z} & \end{aligned}$$

Integrality constraints



P: $\max cx$

$$Ax \leq b$$

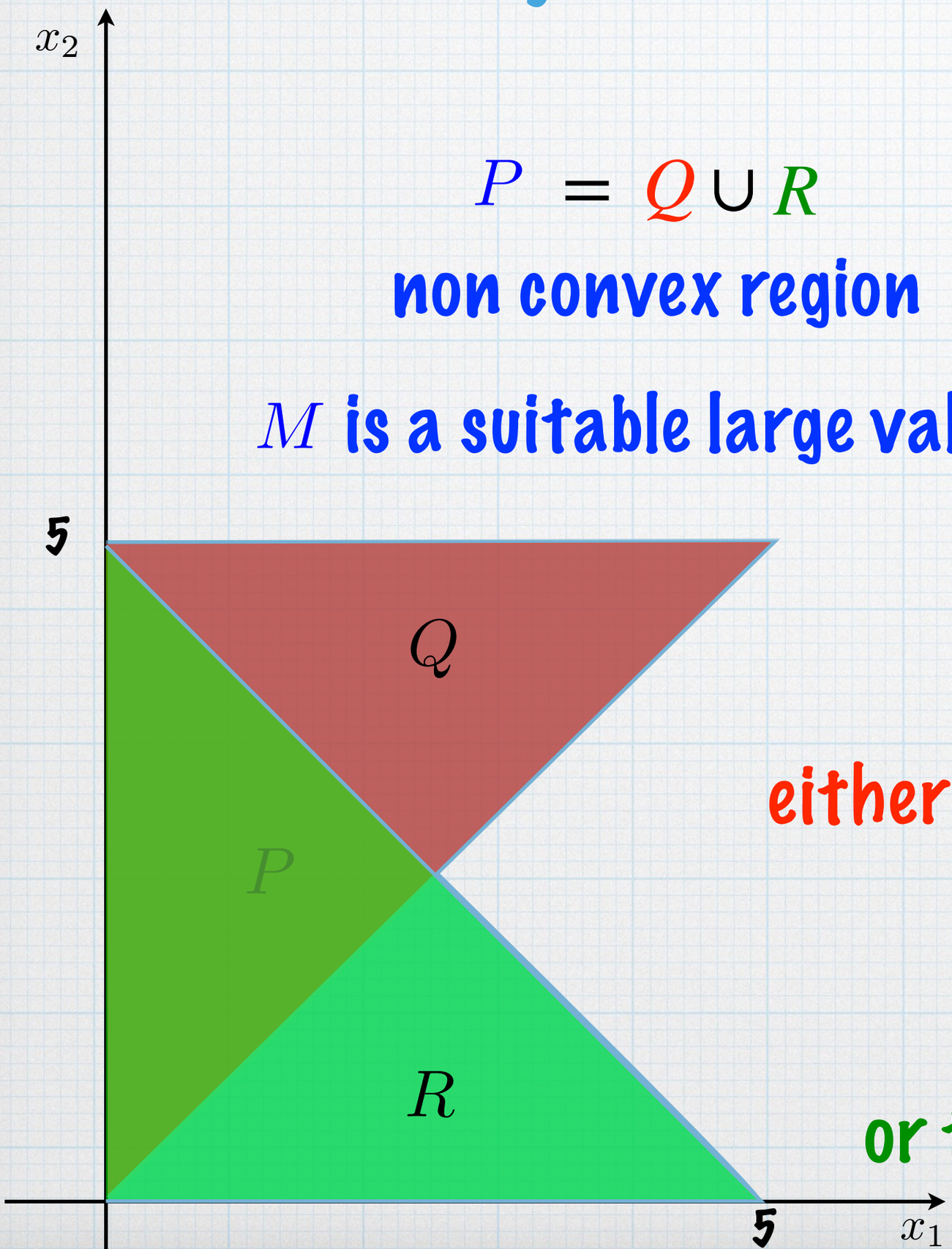
$$x \in \mathbb{Z}^n$$

$$\begin{aligned} \max 30x_D + 50x_W & \\ x_D & \leq 3.5 \\ 2x_W & \leq 11 \\ 3x_D + 2x_W & \leq 16 \\ x_D, x_W & \geq 0 \\ \cancel{x_D, x_W \in \mathbb{Z}} & \end{aligned}$$

Convexity is lost

neglecting the integrality constraints and solving the so called LP relaxation we may obtain a solution with fractional components

Disjunctive constraints



$P = Q \cup R$

non convex region

M is a suitable large value

P :

$$x_1 - x_2 \leq 0 + M(1-y)$$
$$x_1 + x_2 \leq 5 + My$$
$$y \in \{0, 1\}$$

Q : $x_2 \leq 5$

$x_1 \geq 0$

either this $\longrightarrow$ $x_1 - x_2 \leq 0$

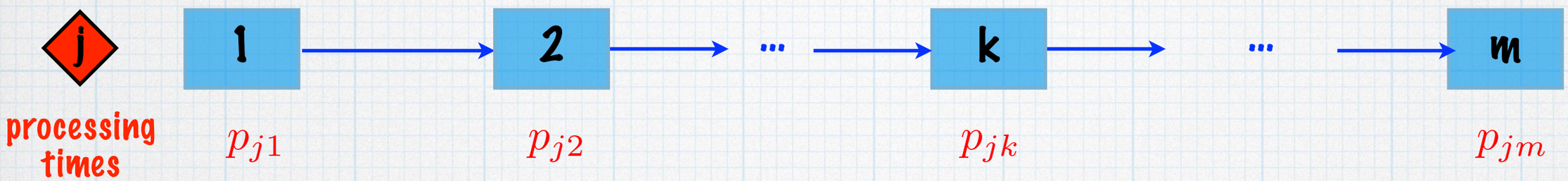
R : $x_1 \geq 0$

$x_2 \geq 0$

or this $\longrightarrow$ $x_1 + x_2 \leq 5$

Scheduling

m machines in a pipeline



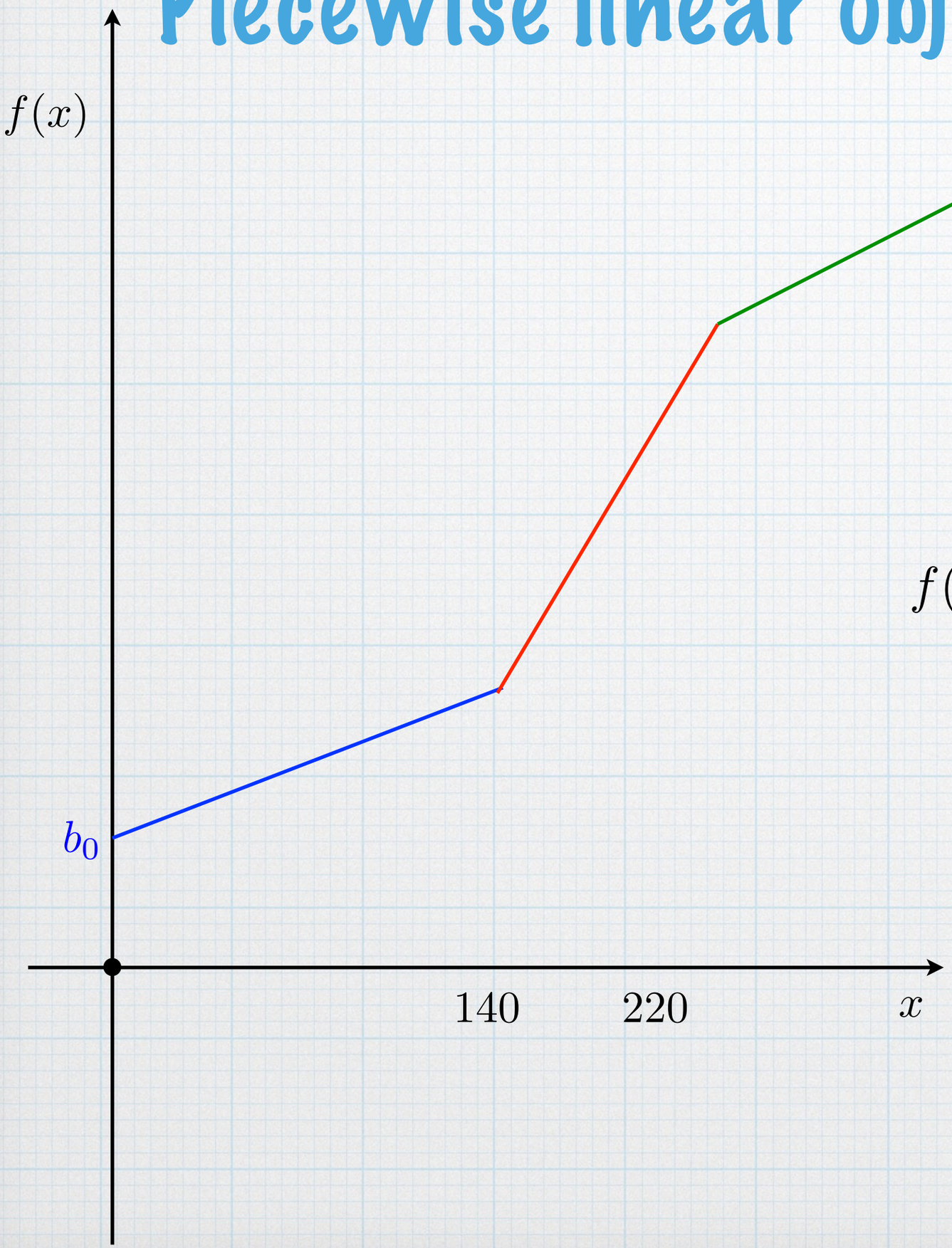
n jobs to be processed by the machines in the pipeline

each machine can process one job at a time

decide for each machine in which order to process the jobs

minimize the ending time of the last machine (makespan)

Piecewise linear objective functions

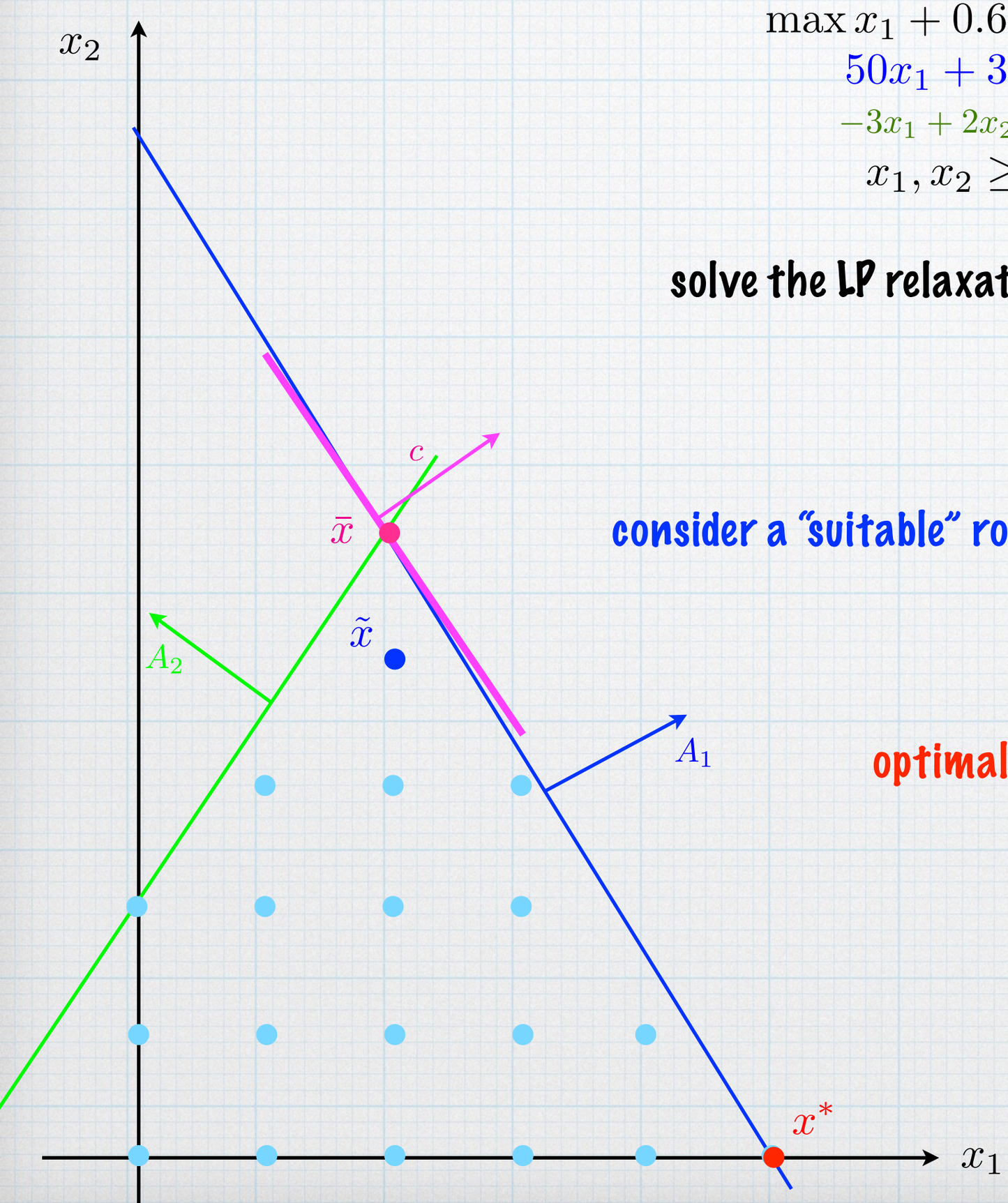


$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ b_0 + 50x & \text{if } 0 < x \leq 140 \\ b_1 + 207x & \text{if } 140 < x \leq 220 \\ b_2 + 127x & \text{if } x > 220 \end{cases}$$

The formulation does matter!

Use LP to solve the problem

$$\max x_1 + 0.64x_2$$
$$50x_1 + 31x_2 \leq 250$$
$$-3x_1 + 2x_2 \leq 4$$
$$x_1, x_2 \geq 0 \quad \text{integer}$$



solve the LP relaxation

fractional coordinates

$$\bar{x} = \begin{bmatrix} 376/193 \\ 950/193 \end{bmatrix} \qquad c\bar{x} = 5.098$$

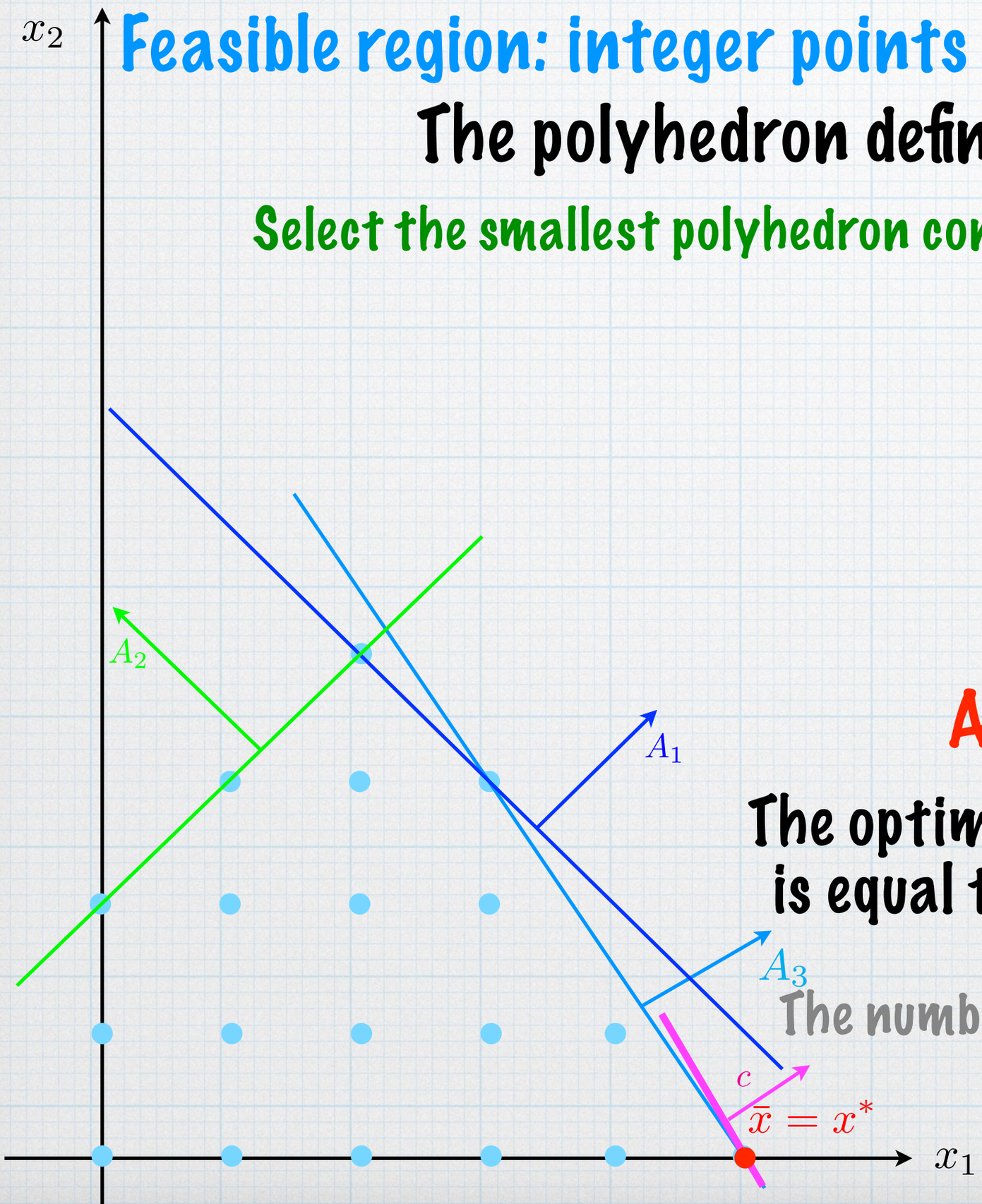
consider a "suitable" rounding

$$\tilde{x} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \qquad c\tilde{x} = 4.56$$

optimal solution

$$x^* = \begin{bmatrix} 5 \\ 0 \end{bmatrix} \qquad cx^* = 5$$

Ideal formulation



Feasible region: integer points contained in a polyhedron

The polyhedron definition is not unique

Select the smallest polyhedron containing all feasible points

$$\max x_1 + 0.64x_2$$
$$x_1 + x_2 \leq 6$$
$$-x_1 + 2x_2 \leq 2$$
$$3x_1 + 2x_2 \leq 15$$
$$x_1, x_2 \geq 0$$

convex hull

All vertices are integer

The optimal solution of the LP relaxation is equal to the optimal integer solution

The number of constraints can be very large

Unimodularity

A is unimodular

if any non singular full rank submatrix has determinant 1 or -1

A is totally unimodular

if any non singular submatrix is unimodular

Consequence: components 0, ±1

Consider problem

$$\min cx$$
$$Ax = b \qquad b \text{ integer}$$

Any basic solution x_B

$$x_B = A_B^{-1} b \quad = \frac{A_B^*}{det(A_B)} b$$ has integer components

Example of TU matrix: node/arc incidence matrix of flow problems

Approximating the convex hull

The complete definition of the convex hull is not necessary

Define the convex hull in the proximity of the optimal solution

$$P: \max cx \quad \begin{array}{l} Ax \leq b \\ x \geq 0 \\ x \in \mathbb{Z}^n \end{array}$$

F

$$P': \max cx \quad \begin{array}{l} Ax \leq b \\ x \geq 0 \end{array}$$

F'

x' optimal solution

Valid inequality: $gx \leq \gamma \quad \forall x \in F$

Cut: $gx' > \gamma$

Cutting plane algorithm

```
optimal←false;
repeat
    solve(P', x')
    if x' is integral then optimal←true
    else generate_cut(g,γ);
        add_cut(P', g, γ);
until optimal
```

Generating valid inequalities

1) The non negative linear combination of valid inequalities is valid

$$A_i x \leq b_i \qquad i = 1, \ldots, m$$

$$\left(\sum_{i=1}^{m} u_i A_i\right) x \leq \sum_{i=1}^{m} u_i b_i \qquad u_i \geq 0, i = 1, \ldots, m$$

2) If $\sum_{j=1}^{n} g_j x_j \leq \gamma$ is valid

$\sum_{j=1}^{n} \lfloor g_j \rfloor x_j \leq \gamma$ is valid

3) If $\sum_{j=1}^{n} \lfloor g_j \rfloor x_j \leq \gamma$ is valid

$\sum_{j=1}^{n} \lfloor g_j \rfloor x_j \leq \lfloor \gamma \rfloor$ is valid

The application of rules 1) 2) and 3) generates Chvàtal valid inequalities

Example

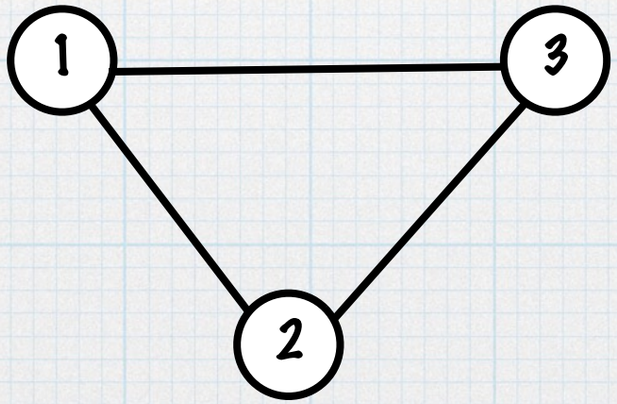
Non bipartite matching

$$G = (N, A)$$

matching $M \subseteq A$

no two arcs are incident in the same node

maximize $|M|$



Gomory cuts

$$P: \quad \begin{array}{l} \min cx \\ \quad Ax = b \\ \quad x \geq 0 \\ \quad x \in \mathcal{Z}^n \end{array}$$

$$\bar{P}: \quad \begin{array}{l} \min cx \\ \quad Ax = b \\ \quad x \geq 0 \end{array}$$

Dual of the asymmetric pair:
more variables than equations

Base B: subset of variable indices $\quad \bar{x}_B \quad \bar{x}_N = 0$

optimal basic solution

constraints of the problem decomposed according to B

$$A_B x_B + A_N x_N = b$$

multiply by A_B^{-1}

$$x_B = \underbrace{A_B^{-1} b}_{\bar{b}} - \underbrace{A_B^{-1} A_N}_{\bar{A}} x_N$$

$\bar{x}_h$ be fractional $\quad$ corresponding to constraint t

$$x_h + \sum_{i \in N} \bar{A}_{tj} x_j = \bar{b}_t$$

is a valid constraint: linear combination of the problem constraints (rule 1)

Gomory cuts (2)

$$x_h + \sum_{i \in N} \lfloor \bar{A}_{tj} \rfloor x_j \leq \bar{b}_t$$ is valid (rule 2)

$$x_h + \sum_{i \in N} \lfloor \bar{A}_{tj} \rfloor x_j \leq \lfloor \bar{b}_t \rfloor$$ is valid (rule 3)

is it a cut?

consider $\bar{x}_B = A_B^{-1} b$
$\bar{x}_N = 0$

the inequality reduces to $\bar{x}_h \leq \lfloor \bar{b}_t \rfloor$

being the optimal basic solution $\bar{x}_h = \bar{b}_t$
and fractional

$$\bar{x}_h > \lfloor \bar{b}_t \rfloor$$

$\bar{x}$ **does not satisfy the inequality**

Solution enumeration

Systematically enumerate al feasible solutions

Evaluate the solutions and pick the best one

Recognize **infeasibility** as soon as possible

Recognize **dominated solutions** and avoid to enumerate them

Example

$$\max 4x_1 + x_2 + 3x_3 + x_4$$

$$5x_1 + 4x_2 + 3x_3 + x_4 \leq 8$$

$$x_i \in \{0, 1\} \quad i = 1, \ldots, 4$$

Organize the enumeration according to a tree

Leaves = solutions
(not all leaves are feasible solutions)

Internal nodes = subproblems

Path from the root to the node identifies the fixed variables

Detect infeasibilities as soon as they occur

Pruning by objective function value

Let us generate heuristically a feasible solution

$x_3 = x_4 = 1 \qquad x_1 = 0 \qquad x_2 = 1$

value = 5

lower bound

we look for a feasible solution having value > 5

Estimate the best solution value of a subproblem without computing all solutions

e.g.: disregard the capacity constraints

upper bound

particular case: the solution of the upper bound is feasible

the upper bound is the optimal value of the subproblem

Scheme of the algorithm (max problem)

```
Procedure Branch&Bound(P, v)
b ← upper_bound(P, x, unfeasible)
if x is feasible for P then return(b)
   else if unfeasible then return(-∞)
      else if b>v then
         Branch(P, x, P1, P2, ..., Pk)
         for i=1...k
            t ← Branch&Bound(Pi,v)
            if t > v then v ← t
```

value of the incumbent solution

optimal solution of the subproblem

detected unfeasibility

the subproblem is still open

update the incumbent solution

How to generate bounds

Consider problem $P : \max\{c(x) : x \in F\}$

Relaxation

$$P' : \max\{c'(x) : x \in F'\}$$

$$F \subseteq F'$$

$$c'(x) \geq c(x) \quad \forall x \in F$$

Properties

$$\max\{c(x) : x \in F\} \leq max\{c'(x) x \in F'\}$$

let $\bar{x}$ be the optimal solution of P'

$$c'(\bar{x}) = c(\bar{x})$$

$$\bar{x} \in F$$

$\bar{x}$ is optimal for P

Types of relaxations

Basic idea:

Relax "complicating constraints" so that the resulting problem is solvable efficiently

In IP relax integrality constraints (LP relaxation)

Omit some constraints

Linear combination of constraints (surrogate relaxation)

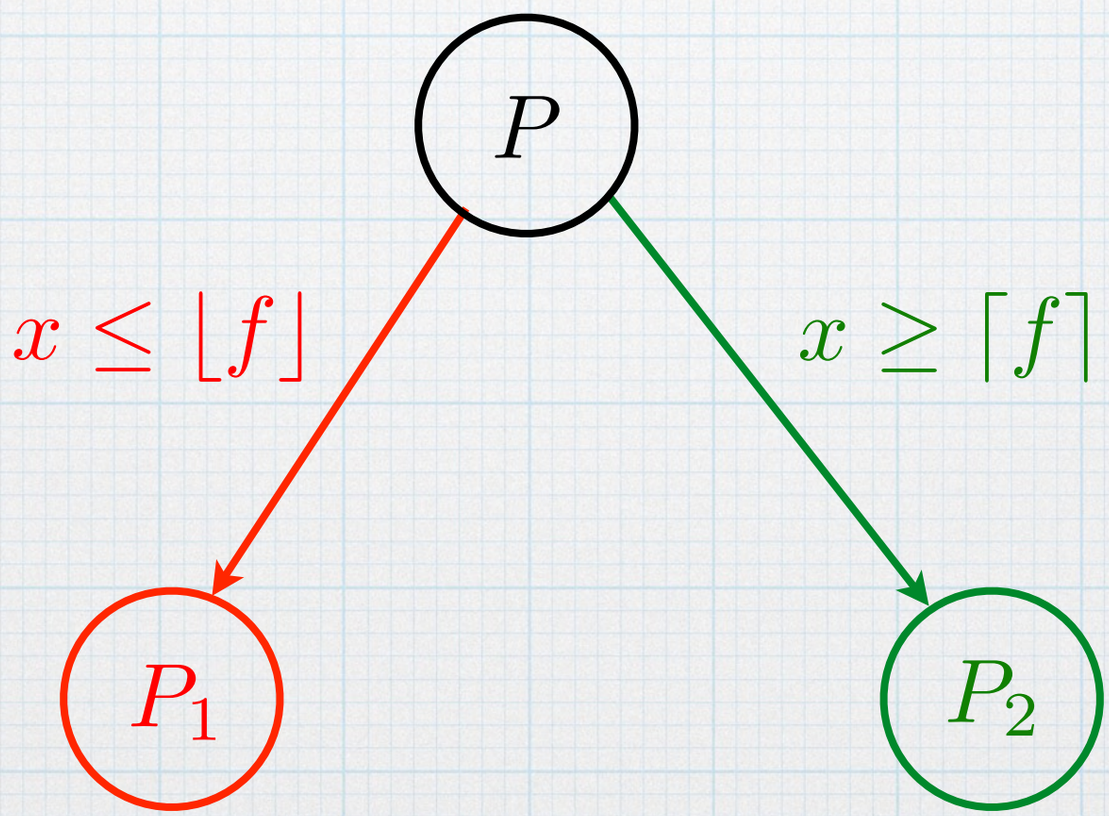
Penalize some constraints in the objective function (Lagrangean relaxation)

Branching strategies

In IP

Consider a variable x

such that $x = f$ is fractional in the LP relaxation



Bipartite branching

ILO of LP section

* Disjunctive constraints
* Piecewise linear functions
* Ideal formulation
* Definitions: valid inequalities and cuts
* How to generate valid inequalities
* Gomory cuts
* The concept of relaxation
* Branch and Bound

Resources Chapter 4

* Section 1
* Section 2
* Section 3